

Storms, Lightning and Cyclones

How rising warm air builds thunderstorms and cyclones, why lightning strikes, and how to stay safe.

From storms to cyclones

Storms and thunderstorms

Warm, moist air rises, cools, and its moisture **condenses** into water droplets forming clouds; these merge and fall as rain. Strong winds with rain make a **storm**.

Inside tall clouds, strong up-and-down winds rub water droplets and ice particles, building up **static charges**. Charges separate (lighter positive ice on top, heavier negative droplets below). When the charge is huge, air's insulation breaks and a sudden flow of charge gives a bright flash — **lightning**. The rapid heating of air makes a loud sound, **thunder**. A storm with lightning and thunder is a **thunderstorm**.

Lightning safety: stay away from tall objects; crouch in a low open area (don't lie flat); avoid metal-rod umbrellas; get out of water; a bus/car is comparatively safe. A **lightning conductor** (metal rod) safely carries the charge to the ground.

Cyclones

Over **warm ocean** water, rising moist air condenses and releases heat, warming the air further so it rises more — creating a very **low-pressure** centre. Surrounding air rushes in, and the **Earth's rotation** makes it spin. This spinning system of clouds, wind and rain is a **cyclone**.

- The calm centre of lowest pressure is the **eye of the cyclone**; around it are strong winds and heavy rain.
- A cyclone **weakens over land** because its supply of warm moist air is cut off.

Where students lose marks

Trap 1 — The eye is calm and lowest pressure. The strongest winds are *around* the eye, not in it.

Trap 2 — Cyclones need warm ocean water + Earth's rotation. Over land the moisture supply is cut off, so they weaken.

Trap 3 — During lightning, crouch — don't lie flat, and avoid tall objects and metal umbrellas.

Lightning: caused by charge separation in clouds; thunder is the sound from rapidly heated, expanding air.

Model answers — write them like this

Example A. Why does a cyclone weaken after it reaches land?

Step 1: A cyclone is fed by warm, moist air rising from the ocean.

Step 2: Over land this moisture supply is cut off.

∴ Without warm moist air, the cyclone loses strength.

Example B. Why is the centre of a cyclone (the eye) calm?

Step 1: The eye is the region of lowest pressure at the centre.

Step 2: The strong winds revolve around it, not within it.

∴ The eye is calm; the surrounding region has strong winds and heavy rain.

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